

Postgres Storage Growth Plan

Q3 2026 storage outlook | Owner: Inez Park, Data Reliability | Next review: 2026-08-17

Decision at a glance

Decision

Add the encrypted volume in August and complete archive partitioning before September close. The governing constraint is migration workspace plus WAL recovery margin, so capacity approval is paired with a restore verification.

Planning peak

58.9 TB

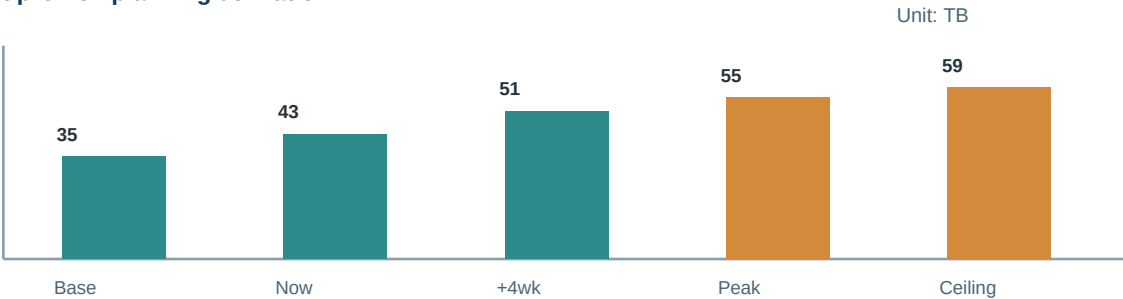
Evidence

High; partition inventory and WAL trends agree

Scope and assumptions

Storage planning covers checkout ledger partitions, index bloat, WAL retention, and synthetic audit snapshots. Growth assumes 1.9 TB monthly ledger expansion, a one-time 3.4 TB index rebuild, and unchanged compression ratios. Row-level data is not included; the report uses aggregate synthetic capacity values only.

Forecast profile - planning utilization



Capacity signals

Measure	Current	Planning limit	Decision signal
Used primary storage	42.8 TB	58.9 TB	Provision at 52 TB to leave migration working space.
Index bloat	19%	< 24%	Rebuild partition indexes before bloat consumes reserve.
WAL backlog	1.6 TB	< 3.0 TB	Backlog over 3 TB delays point-in-time recovery.
Archive transfer	680 MB/s	>= 540 MB/s	Slow transfer moves cold-tier schedule earlier.

Plan, controls, and ownership

Action plan

Action	Due	Owner	Acceptance evidence
Volume add	Aug 01	Database platform	Attach one encrypted 24 TB volume.
Archive partition	Aug 08	Data engineering	Move closed monthly partitions to cold tier.
Bloat maintenance	Aug 13	DBA rotation	Rebuild two largest synthetic indexes.
Recovery check	Aug 17	Inez Park	Restore a capacity sample within the RTO envelope.

Risk register

Risk	Likelihood	Impact	Response
Archive throughput drop	Medium	Medium	Throttle compaction before consuming primary reserve.
Rebuild amplification	Medium	High	Run one partition at a time with WAL watch.
Retention exception	Low	High	Require data-governance approval for any extension.

Decision rationale

Add the encrypted volume in August and complete archive partitioning before September close. The governing constraint is migration workspace plus WAL recovery margin, so capacity approval is paired with a restore verification.

Selective cross-references

Review alongside: Observability retention capacity and Checkout demand forecast. These linked plans are relevant because they share a limiting dependency or coordinated operating window.

Model notes: all values are synthetic planning figures. No customer records, production identifiers, secrets, or routable network addresses are included. Northstar Transit Cloud | Storage estimates exclude customer content